

# Logistic-Bernoulli-Model

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## Load library

```
library(coda)
library(rjags)
```

```
## Linked to JAGS 4.3.1
```

```
## Loaded modules: basemod,bugs
```

```
library(knitr)
library(purrr)
library(tidyr)
library(ggplot2)
```

## set seed

```
set.seed(123)
```

## Load Data

```
df <- read.csv('./drug200.csv')
head(df)
```

```
##   Age Sex   BP Cholesterol Na_to_K  Drug
## 1  23  F  HIGH         HIGH  25.355 DrugY
## 2  47  M  LOW         HIGH  13.093 drugC
## 3  47  M  LOW         HIGH  10.114 drugC
## 4  28  F NORMAL        HIGH   7.798 drugX
## 5  61  F  LOW         HIGH  18.043 DrugY
## 6  22  F NORMAL        HIGH   8.607 drugX
```

## Feature Encoding

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```
df$Sex <- as.numeric(factor(df$Sex))
df$BP <- as.numeric(factor(df$BP))
df$Drug <- as.numeric(factor(df$Drug))
df$Cholesterol <- replace(df$Cholesterol, df$Cholesterol=='HIGH', 1)
df$Cholesterol<- replace(df$Cholesterol, df$Cholesterol=='NORMAL', 0)
```

## Split Train-Test Data

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```
## 70% of the sample size
smp_size <- floor(0.7 * nrow(df))

## set the seed to make your partition reproducible
set.seed(123)
train_ind <- sample(seq_len(nrow(df)), size = smp_size)

train <- df[train_ind, ]
test <- df[-train_ind, ]
```

## Set Input Variables

```
Y <- as.matrix(train$Cholesterol)
X <- train[, cbind('Age', 'Sex', 'BP', 'Na_to_K', 'Drug')]
names <- cbind('Age', 'Sex', 'BP', 'Na_to_K', 'Drug')
n <- length(Y)
p <- ncol(X)

burn <- 500
iters <- 2000
chains <- 2
```

## Delete Missing Value

```
junk <- is.na(rowSums(X))
Y <- Y[!junk]
X <- X[!junk,]
```

## Scale X

```
X <- as.matrix(scale(X))
```

## Modelling

# Make Model String

```
mod <- textConnection("model{
# Likelyhood
for(i in 1:n){
  Y[i] ~ dbern(pi[i])
  logit(pi[i]) <- alpha + mu[i]
  mu[i] <- inprod(X[i,], beta[])
  like[i] <- dbin(Y[i],pi[i],1) # For WAIC computation
}

# Prior
alpha ~ dnorm(0,0.01)
for(j in 1:p){
  beta[j] ~ dnorm(0,0.01)
}
}")
```

## Fit and Compile Model

```
data <- list(Y=Y,X=X,n=n,p=p)
model <- jags.model(mod,data = data, n.chains=chains,quiet=TRUE)
update(model, burn)
samps <- coda.samples(model, variable.names=c("like"),
n.iter=iters, n.thin = 5)
```

## DIC and WAIC

```
# Compute DIC
DIC <- dic.samples(model,n.iter=iters,n.thin = 5)
# Compute WAIC
like <- rbind(samps[[1]],samps[[2]]) # Combine samples from the two chain
fbar <- colMeans(like)
Pw <- sum(apply(log(like),2,var))
WAIC <- -2*sum(log(fbar))+2*Pw
DIC
```

```
## Mean deviance: 192.6
## penalty 6.096
## Penalized deviance: 198.7
```

```
WAIC
```

```
## [1] 199.049
```